

MIDDLE PENINSULA, VA, USA

WMO: 720285

Lat: 37.521N

Lon: 76.765W

Elev: 24

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03734

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.8	21.3	-3.7	4.5	26.2	2.7	6.3	29.1	22.3	51.6	18.8	49.7	3.6	320	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	95.0	78.0	92.0	76.7	90.2	75.8	81.1	89.7	79.7	87.4	78.4	85.8	6.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	151.3	84.8	77.4	142.9	83.5	76.5	138.2	82.4	44.6	88.6	43.1	87.1	41.9	86.2	97.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.5	15.1	12.5	DB	8.6	98.5	8.1	2.6	2.8	100.4	-1.9	101.9	-6.4	103.4	-12.3	105.3	(4)
(5)			WB	7.3	82.4	7.2	3.4	2.1	84.8	-2.1	86.8	-6.1	88.7	-11.3	91.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.9	39.4	41.8	48.6	59.3	68.4	75.6	79.9	78.0	72.2	61.3	49.9	43.2		
	DBStd	16.49	10.41	9.87	9.92	8.85	7.75	5.99	4.60	4.74	6.09	8.53	8.94	9.69		
	HDD50	1177	358	268	153	18	0	0	0	0	0	9	110	261		
	HDD65	3564	795	650	515	216	58	3	0	0	12	174	461	680		
	CDD50	4792	28	39	110	298	569	768	928	869	667	361	105	50		
	CDD65	1704	0	1	7	46	162	321	463	404	229	60	7	4		
	CDH74	16339	1	8	75	485	1356	3148	4899	3940	1875	440	84	28		
	CDH80	6379	0	0	9	128	446	1246	2168	1604	623	97	46	12		
(14) Wind	WSAvg	4.8	5.2	5.6	6.0	6.2	5.2	4.5	4.2	3.7	4.0	4.4	4.4	4.5		
(15) Precipitation	PrecAvg	44.9	3.4	2.7	4.2	3.3	3.8	3.7	4.6	4.4	4.6	3.5	3.1	3.5		
	PrecMax	64.9	6.3	6.2	8.7	6.8	7.8	5.9	8.5	13.6	15.9	7.1	8.9	8.9		
	PrecMin	34.6	1.2	0.6	0.2	1.3	0.9	1.6	1.3	1.3	0.2	0.0	0.2	1.1		
	PrecStd	7.4	1.3	1.5	1.8	1.3	1.7	1.3	2.2	2.8	3.6	1.9	2.0	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.1	74.2	81.2	88.4	91.3	97.2	99.5	97.3	93.4	87.8	83.9	75.6		
		MCWB	61.7	62.4	62.6	67.2	72.7	77.8	77.9	80.2	73.9	72.2	75.0	65.8		
	2%	DB	66.0	69.7	75.3	83.7	88.4	92.8	95.1	93.0	90.0	82.2	73.2	68.8		
		MCWB	59.5	58.3	60.3	65.4	71.7	76.7	78.5	77.3	74.8	71.0	64.0	64.1		
	5%	DB	61.6	64.1	71.0	80.0	84.9	90.0	92.4	90.6	86.4	79.1	69.9	64.0		
		MCWB	55.4	55.0	58.1	64.1	70.6	75.2	77.8	76.6	73.0	68.8	62.3	60.1		
	10%	DB	56.6	59.0	66.0	75.3	81.8	87.4	90.2	88.2	83.6	75.2	65.0	59.1		
		MCWB	50.6	51.0	56.5	62.2	69.3	74.2	76.9	75.5	71.7	67.5	59.0	54.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.5	64.7	67.1	71.5	76.8	81.0	83.1	82.9	79.3	77.0	74.1	69.3		
		MCDB	68.8	71.5	73.8	79.2	86.7	89.3	93.2	92.7	85.5	81.5	79.6	71.7		
	2%	WB	60.7	61.5	64.0	69.3	74.7	79.1	81.2	81.0	77.5	74.7	66.7	64.4		
		MCDB	65.2	66.2	71.1	78.3	82.9	87.1	90.5	88.8	83.3	79.4	70.7	67.5		
	5%	WB	56.6	57.0	61.2	67.0	73.0	77.6	79.9	79.5	76.2	72.4	63.6	60.7		
		MCDB	61.5	63.3	68.1	76.4	80.8	86.3	88.3	86.3	81.9	76.5	68.0	63.9		
	10%	WB	51.4	51.7	57.8	64.8	71.4	76.3	78.5	78.0	74.8	69.2	60.3	55.5		
		MCDB	55.2	57.3	64.9	72.7	78.7	84.2	86.4	84.6	80.6	74.1	64.0	58.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.4	20.2	21.4	22.7	20.1	19.5	18.3	18.5	19.4	20.4	21.4	19.1		
		MCDBR	24.6	26.1	26.8	26.4	21.8	21.4	20.1	20.1	22.2	22.7	23.7	22.8		
		MCWBR	19.1	17.4	15.4	11.7	8.9	8.4	7.5	7.6	9.0	11.8	15.1	18.4		
	5% WB	MCDBR	23.3	24.8	22.6	22.3	18.7	18.6	18.0	17.4	16.9	17.3	19.7	22.1		
		MCWBR	20.2	19.2	16.1	11.8	9.2	8.1	7.9	8.0	8.5	10.3	15.7	19.3		
(40) Clear-Sky Solar Irradiance	taub		0.305	0.317	0.348	0.392	0.431	0.485	0.515	0.511	0.410	0.354	0.322	0.309		
	taud		2.532	2.490	2.439	2.327	2.249	2.144	2.080	2.076	2.344	2.500	2.544	2.555		
	Ebn at Noon		282	291	291	282	271	256	247	244	266	274	272	272		
	Edh at Noon		24	29	33	40	43	49	51	50	36	29	24	23		
(44) All-Sky Solar Radiation	RadAvg		728	970	1292	1631	1853	2011	1919	1733	1404	1101	824	645		
	RadStd		59	106	106	120	174	141	94	85	133	128	69	53		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (75 neighbors)		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+284	+180

Nomenclature: See separate page